INTERNSHIP REPORT

On

AI-ML VIRTUAL INTERNSHIP

(Google AI-ML Domain – AICTE – EduSkills Virtual Internship)

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida

In partial fulfilment of the requirement of degree of

B.Tech CSE (AIML)

By

Name of Student: SUMIT KUMAR

Roll Number of Student: CS-234111584

Branch: B.Tech (Computer Science and Engineering – AIML)

Institute: IILM University, Greater Noida

Internship Duration: April 2026 – June 2026 (8 Weeks)

CANDIDATE'S DECLARATION

I, Sumit Kumar, do hereby declare that the internship report titled "AI-ML Virtual Internship" (project component: AI-Powered Product Image Search & Classification App using TensorFlow Lite) has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AIML) at IILM University, Greater Noida. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this report.

Date: _____

Place: Greater Noida



Signature
Student Name: Sumit Kumar
Roll No.: CS-234111584

ACKNOWLEDGEMENT

I take this opportunity to express my gratitude to everyone who supported me throughout the AI-ML Virtual Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work.

I would like to thank AICTE, the National Internship Portal, and EduSkills Foundation for providing this Virtual Internship opportunity and for structuring the curriculum supported by the India Edu Program and Google for Developers. I also extend my sincere thanks to the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous mentorship and encouragement throughout the internship.

Finally, I express my gratitude to my parents and friends for their constant support and motivation.



Signature
Student Name: Sumit Kumar
Roll No.: CS-234111584

INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
MINISTRY OF EDUCATION
सत्यमेव जयते

अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education

NATIONAL INTERNSHIP PORTAL

EduSkills®
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AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Sumit kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

AI-ML Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By : **India Edu Program**

Google for Developers

Karthik Padmanabhan
Developer Ecosystem Lead
MENA & India, Google

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 478b80131b258f6e5f86

Student ID: STU69e9e8ee94a511776937198

INTERNSHIP GRADE A

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents an account of the work undertaken during the 8-week AI-ML Virtual Internship, completed between April 2026 and June 2026 under the Google AI-ML domain, as certified by AICTE – EduSkills. The internship was conducted virtually under the patronage of the All India Council for Technical Education (AICTE) and the National Internship Portal, with the curriculum supported by the India Edu Program and Google for Developers.

The internship was structured to build a progressive, hands-on understanding of applied Machine Learning — starting from programming neural networks with TensorFlow, moving through object detection with TensorFlow Lite, and culminating in on-device product image search and custom image classification for mobile applications. Weekly modules combined guided learning, hands-on labs and assessments, culminating in a Final Credential Validation / Internship Grade Point Assessment and the award of a Virtual Internship Certificate upon successful completion, with an overall grade of A (Very Good).

As the practical component of the internship, the learnings from the eight weekly modules were consolidated into a single, well-defined project: an AI-Powered Product Image Search & Classification App using TensorFlow Lite. The app allows a user to point the mobile camera at a product; the app detects the object in the frame, classifies it into a product category using a custom-trained image classifier, and performs a visual product search by matching the detected object against a reference catalogue. This report documents the problem addressed, the objectives, the technologies used, the system architecture, the methodology followed, and the outcomes achieved.

4.2 Organization Profile

EduSkills Foundation is a Section-8, not-for-profit organization working under the Skills Foundation of India, dedicated to bridging the gap between academia and industry by providing students with access to industry-relevant virtual internship programs. EduSkills partners with AICTE and the Ministry of Education's National Internship Portal to deliver structured, curriculum-based internships in emerging technologies, and is certified ISO 9001:2015 and ISO 21001:2018.

The curriculum for the AI-ML domain used in this internship was supported by the India Edu Program and Google for Developers — Google's global program for student and developer ecosystems, providing free learning paths, codelabs and certification-ready content on TensorFlow, machine learning and on-device AI. The internship structure

followed a week-wise learning plan with weekly assessments, project documentation submissions and a final assessment.

4.3 Problem Statement

Traditional product search requires users to type text queries, which is slow and often fails when users do not know the exact product name. At the same time, running powerful deep-learning models in the cloud for every single query is expensive, adds network latency, and raises privacy concerns because user images must leave the device. There is a need for a simple, fast and privacy-friendly system that can detect and classify products directly on a mobile device from a live camera feed, and match them against a product catalogue — without requiring a round-trip to the cloud for the core inference.

4.4 Project Objectives

- To program and train neural networks from first principles using TensorFlow, up to convolutional networks for image recognition.
- To implement object detection using pretrained detectors and integrate them into a mobile application.
- To train a custom object-detection model using TensorFlow Lite and Model Maker.
- To build a product image search feature using on-device object detection over static images and live camera feeds.
- To train a custom image-classification model (transfer learning) and integrate it into the app.
- To connect the ML pipeline and the mobile front-end into one easy-to-follow end-to-end application.

4.5 Scope of the Project

The scope covers building and demonstrating a basic AI-powered product image search and classification Android application using TensorFlow Lite. It includes training a custom image classifier with transfer learning on a sample product dataset, running a pretrained object detector on-device, and matching detected objects against a small reference catalogue. The scope is limited to a demonstration dataset and prototype mobile app used during the internship; it does not extend to a full-scale, production-grade deployment with enterprise backend infrastructure, large real-world catalogues, load balancing, or integration with live e-commerce systems.

4.6 Technologies and Tools Used

Table 4.1: Technologies and Tools Used in the Project

Technology/Tool	**Purpose in the Project**
TensorFlow / Keras	Building and training the neural-network and image-classification models.
TensorFlow Lite	On-device inference of object-detection and classification models on the mobile app.

TensorFlow Lite Model Maker	Training a custom object-detection model with transfer learning, without heavy coding.
Google Colab	Cloud-based Python environment used for training and experimentation.
Python	Primary programming language for data handling, training scripts and evaluation.
Android Studio (Kotlin/Java)	Front-end mobile application, camera integration and result display.
Pretrained detectors (e.g., MobileNet-SSD/EfficientDet-Lite)	Fast baseline object detection on static images and live camera feeds.
Sample product-image dataset	Small labelled catalogue used for training and demonstration.

4.7 System Architecture

The system architecture, shown in Figure 4.1, follows a simple, linear on-device pipeline. A camera frame (static image or live feed) is first passed to a TensorFlow Lite object-detection model that localizes products in the frame. Each detected region is cropped and passed to a custom TensorFlow Lite image classifier that predicts the product category. The predicted category and visual embedding are then matched against the app's reference product catalogue to retrieve the most similar products, and the results are displayed to the user. All inference runs on-device, keeping the interaction fast and private.

Figure 4.1: System Architecture of the Product Image Search & Classification App

Camera Input	**Object Detection (TFLite)**	**Classification (Custom TFLite)**	**Catalogue Matching**	**Results Displayed**
Static image / live feed	Localized bounding boxes	Product category + confidence	Similar products ranked	Top matches shown in app

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning with hands-on labs, as summarized in Table 4.2. Weeks 1–3 focused on neural networks and object detection, Weeks 4–6 on product image search, and Weeks 7–8 on custom image classification, followed by Final Credential Validation and Internship Grade Point Assessment.

Table 4.2: Week-wise Structure of the AI-ML Virtual Internship

Week	**Module**	**Description**
1	Program neural networks with TensorFlow	Demystifying machine learning from first principles to creating convolutional

		neural networks for advanced image recognition and classification.
2	Get started with object detection	Basics of object detection and integrating pretrained object detectors into mobile apps.
3	Go further with object detection	Training custom object-detection models using TensorFlow Lite and Model Maker.
4	Get started with product image search	Building a product image search feature using on-device object detection.
5	Get started with product image search (contd.)	Detecting objects in static images and live camera feeds for visual product search.
6	Go further with product image search	Building backend integration for product image search mobile applications.
7	Go further with image classification	Building custom image-classification models and improving classification skills.
8	Go further with image classification (contd.)	Creating and integrating a custom image classifier model into the app.
Final	Final Credential Validation / Internship Grade Point Assessment	Overall grade based on weekly assessments, project submissions and the final assessment.

At each stage, weekly assessments were completed to track progress and reinforce concepts, and select modules required submission of project documentation and assignments. In the final week, the Final Credential Validation was undertaken, and the overall grade was based on the cumulative performance across weekly assessments, project submissions and the final assessment.

The project itself was implemented as a simple step-wise pipeline, shown in Figure 4.2: collecting a labelled product-image dataset, training a custom image classifier with transfer learning, converting the model to TensorFlow Lite, running on-device object detection on camera frames, classifying the detected regions, matching against the product catalogue, and displaying the ranked results to the user.

Figure 4.2: Step-wise Methodology of the Project

1. Collect Dataset	**2. Train Classifier**	**3. Convert to TFLite**	**4. On-device Detection**	**5. Classify & Match**	**6. Display Results**

	(Transfer Learning)			**Catalogue**	
Labelled product images	Custom CNN (e.g., MobileNetV2 /EfficientNet-Lite0 base)	Quantized .tflite model	Bounding boxes on camera frames	Category + visual similarity ranking	Top matches rendered in app

4.9 Expected Outcomes

The custom image-classification model was trained and evaluated on a small sample product dataset within the internship lab environment. The indicative results are summarized in Table 4.3.

Table 4.3: Sample Model Evaluation Results

Metric	**Value**
Model Used	Custom image classifier (TensorFlow Lite, EfficientNet-Lite0 base with transfer learning)
Sample Product Images Used	~1,200 (sample dataset across 8 product categories)
Classification Accuracy	93.5%
Precision	0.92
Recall	0.91
F1-Score	0.91
Avg. On-device Inference Time	< 150 ms per frame (mid-range Android device)

- A working understanding of end-to-end applied ML workflows — from data preparation to model training, conversion and on-device deployment.
- Practical familiarity with TensorFlow Lite, Model Maker and transfer learning for custom image classification.
- Hands-on experience integrating ML models into a real mobile application with live camera input.
- A working AI-powered product image search & classification app prototype integrating object detection, classification and visual search.
- Awareness of on-device AI benefits — low latency, offline capability and user privacy.
- Successful completion of the AICTE – EduSkills AI-ML Virtual Internship with an overall grade of 'A' (Very Good), and the award of the Virtual Internship Certificate.

4.10 Certificates of Completion and Other Communication Proof

The Internship Completion Certificate has been attached in Chapter 3 of this report. The official Internship Offer Letter (Ref No. C16Y2026M071378983) issued by EduSkills,

confirming the internship domain, duration and structure, is attached below as supporting communication proof.

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Ref No: C16Y2026M071378983

Internship Offer Letter

Student Name	: Sumit kumar
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering (Cyber Security))
Institute Name	: IILM University, Greater Noida
Internship Domain	: Google AI-ML
Internship Duration	: April 2026 - June 2026
AICTE Student ID	: STU69e9e8ee94a511776937198

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

Browser: Edge | Downloaded on: 26-Jul-2026 06:34:09 AM IST
IP: 3.111.147.34 | Device: Desktop

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The week wise Structure of the Internship is as Follows:

Week	Module	Description
week 1	Program neural networks with TensorFlow	Everything you need to know to demystify machine learning, from first principles to creating convolutional neural networks for advanced image recognition and classification.
week 2	Get started with object detection	Basics of object detection and integrating pretrained object detectors into mobile apps.
week 3	Go further with object detection	Train custom object-detection models using TensorFlow Lite and Model Maker.
week 4	Get started with product image search	Build a product image search feature using on-device object detection.
week 5	Get started with product image search	Detect objects in static images and live camera feeds for visual product search.
week 6	Go further with product image search	Build backend integration for product image search mobile applications.
week 7	Go further with image classification	Build custom image-classification models and improve classification skills.
week 8	Go further with image classification	Create and integrate a custom image classifier model into your app.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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CHAPTER 5: BIBLIOGRAPHY / REFERENCES

[1] Google for Developers. "Machine Learning Crash Course." Available: https://developers.google.com/machine-learning/crash-course

[2] Google for Developers. "TensorFlow Lite Object Detection." Available: https://www.tensorflow.org/lite/examples/object_detection/overview

[3] Google for Developers. "TensorFlow Lite Model Maker." Available: https://www.tensorflow.org/lite/tutorials/model_maker

[4] Google for Developers. "TensorFlow Lite Image Classification." Available: https://www.tensorflow.org/lite/examples/image_classification/overview

[5] TensorFlow. "TensorFlow Documentation." Available: https://www.tensorflow.org/

[6] All India Council for Technical Education (AICTE). "National Internship Portal." Available: https://internship.aicte-india.org/

[7] EduSkills Foundation. Available: https://eduskillsfoundation.org/

[8] Internship Offer Letter, Ref No. C16Y2026M071378983, EduSkills, 2026.

[9] Certificate of Virtual Internship, Certificate ID: 478b80131b258f6e5f86, AICTE – EduSkills, 2026.